The Six-bar Mechanisms

Six-bar linkages with one degree of freedom have two variants – the Watt's form and the Stephenson's form. Five different mechanisms are derivable by kinematic inversion of the Watt and Stephenson 6-bar chains, namely the Watt I & II, and Stephenson-I, II & III linkages. Out of these, only three – Watt-I, and Stephenson-I & II – produce six-bar coupler curves, which have been found to be of degrees 14, 14 and 16 respectively. The rest of the six-bar linkages generate four-bar coupler curves.

MechAnalyzer includes the Watt-I & II, and Stephenson-I & III mechanisms.

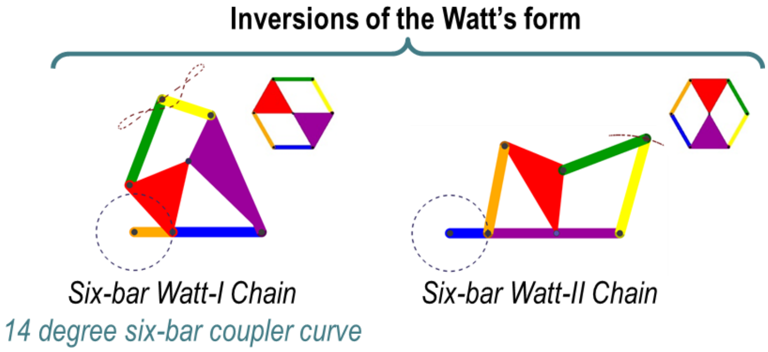

Figure 1: *Inversions of Watt's form of the six-bar linkage.*

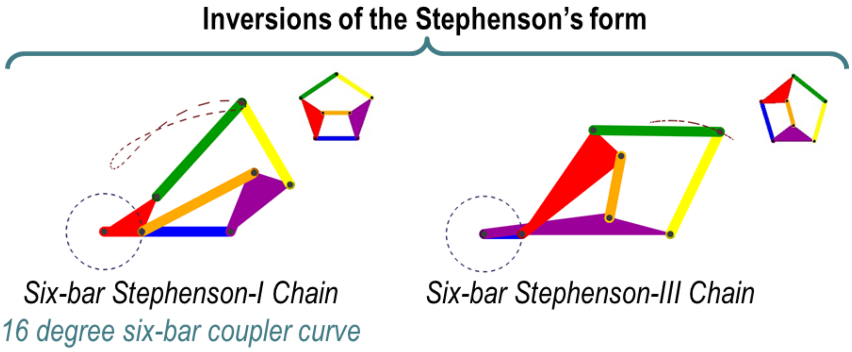

Figure 2: *Inversions of Stephenson's form of the six-bar linkage.*

Applications

1. ***Thread take-up lever in a sewing machine***. Provides superior thread displacement curve than a four-bar linkage, and lesser noise than a cam mechanism.
2. ***In mechanical calculators***. Example: A patented linkage closely approximating the logarithmic function. (US patent 2340350, Antonin Svoboda)